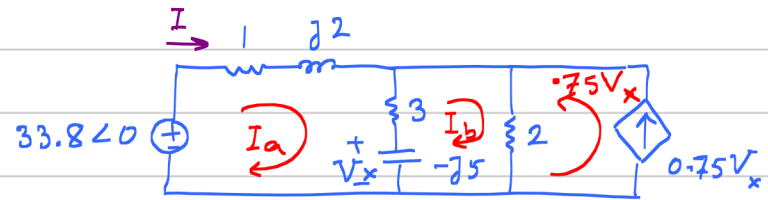




Ex. Use mesh analysis to find I.



$$(1+j2)I_a + (3-j5)(I_a-I_b) = 33.8\angle 0 \quad (1)$$

$$(3-j5)(I_b-I_a) + 2 \left[I_b + 0.75(-j5)(I_a-I_b) \right] = 0 \quad (2)$$

$$(1) \Rightarrow (4-j3)I_a - (3-j5)I_b = 33.8$$

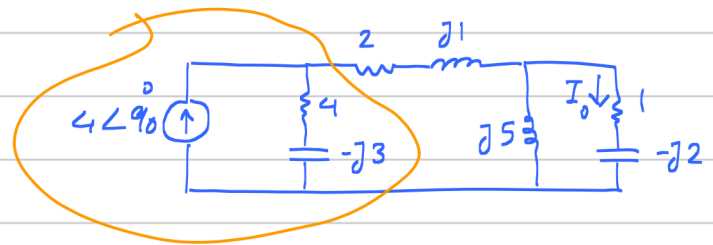
$$(2) \Rightarrow (-3-j2.5)I_a + (5+j2.5)I_b = 0$$

$$I_b = \frac{3+j2.5}{5+j2.5} I_a = 0.698\angle 13.25^\circ I_a$$

$$(4-j3)I_a - (3-j5)(0.698\angle 13.25^\circ)I_a = 33.8$$

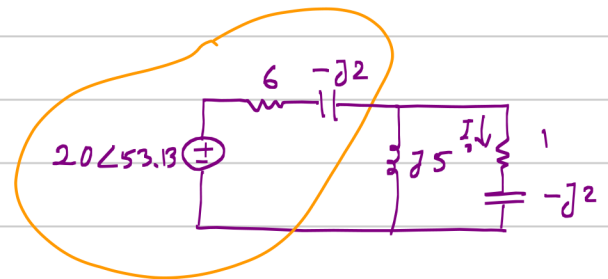
$$I_a = 29.1\angle 3.9^\circ A \rightarrow I = I_a = 29.1\angle 3.9^\circ A$$

Ex. Find I_o using source transformation.



$$(4\angle 90^\circ)(4-j3) = 20\angle 53.13^\circ \text{ V}$$

$$4-j3 + 2+j1 = 6-j2 \Omega$$



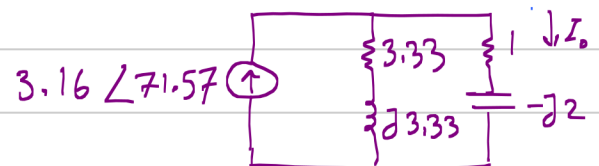
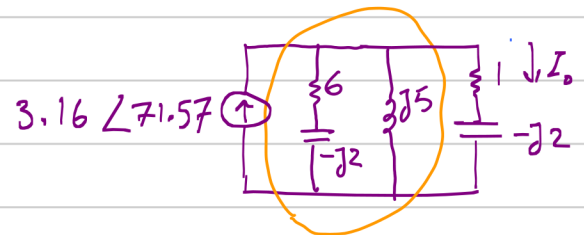
$$\frac{20\angle 53.13^\circ}{6-j2} = 3.16\angle 71.57^\circ$$

$$j5 \parallel (6-j2) = 3.33 + j3.33$$

C.D.R:

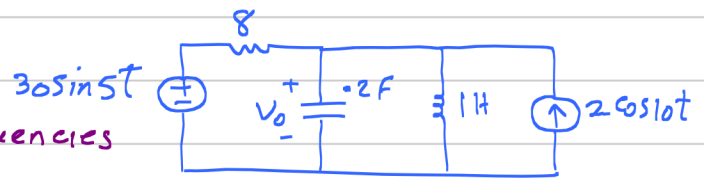
$$I_o = (3.16\angle 71.57^\circ) \frac{3.33 + j3.33}{4.33 + j1.33}$$

$$= 3.29\angle 99.5^\circ \text{ A}$$



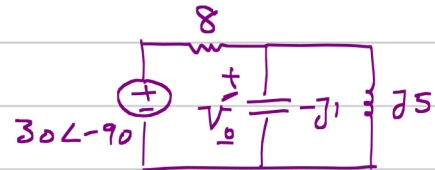
Ex. Calculate $V_o(t)$, using superposition.

The two sources are not at the same frequencies



at $\omega_1 = 5$: $Z_L = j\omega_1 L = j(5)(1) = j5 \Omega$

$Z_C = \frac{-j}{\omega_1 C} = -j1 \Omega$

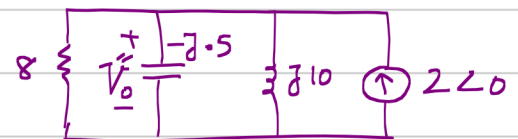


$$V_o' = (30 \angle -90) \frac{-j1 \parallel j5}{8 + (-j1 \parallel j5)} = 4.63 \angle -171.1^\circ$$

$$V_o'(t) = 4.63 \cos(5t - 171.1^\circ)$$

at $\omega_2 = 10$: $Z_L = j\omega_2 L = j10 \Omega$

$Z_C = \frac{-j}{\omega_2 C} = -j0.5 \Omega$



$$\frac{1}{Z_{eq}} = \frac{1}{8} + \frac{1}{-j0.5} + \frac{1}{j10} = \frac{5 + j76}{40}$$

$$Z_{eq} = 0.525 \angle -86.235^\circ \Omega$$

$$V_o'' = (2 \angle 0) (0.525 \angle -86.235) = 1.05 \angle -86.24^\circ \text{ V}$$

$$V_o''(t) = 1.05 \cos(10t - 86.24^\circ)$$

$$V_o(t) = V_o'(t) + V_o''(t) = 4.63 \cos(5t - 171.1^\circ) + 1.05 \cos(10t - 86.24^\circ) \text{ V}$$

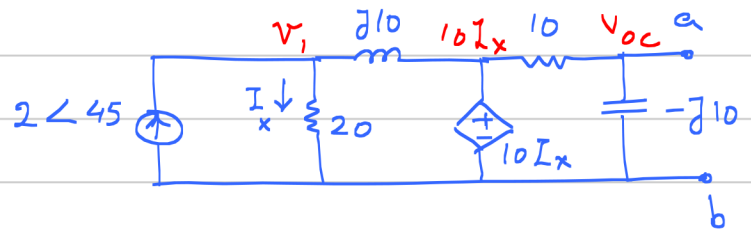
Note: Since the circuit is energized by two sources at different

frequencies, superposition is the only technique that can be used.

After finding the individual responses, add them only in time domain.

Ex. Find the Thevenin equivalent circuit.

$$V_{Th} = V_{oc} = V_c$$



$$\text{KCL: } -2\angle 45 + \frac{V_1}{20} + \frac{V_1 - 10(\frac{V_1}{20})}{j10} = 0$$

$$\frac{V_1(1-j1)}{20} = 2\angle 45 \Rightarrow V_1 = \frac{40\angle 45}{\sqrt{1^2+1^2} \angle \tan^{-1} \frac{-1}{1}}$$

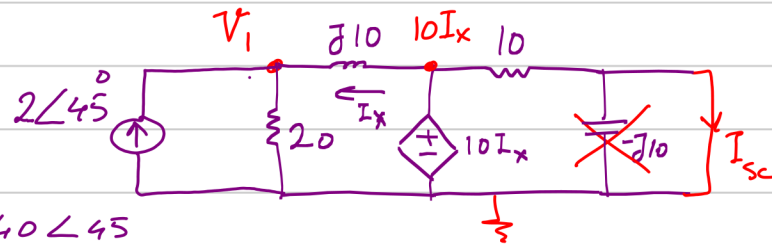
$$V_1 = \frac{40\angle 45}{\sqrt{2} \angle -45} = 20\sqrt{2} \angle 90$$

$$\text{KCL: } \frac{V_{oc}}{-j10} + \frac{V_{oc} - 10(\frac{V_1}{20})}{10} = 0$$

$$\Rightarrow V_{Th} = V_{oc} = \boxed{10\angle 45^\circ} \text{ V}$$

$$Z_{Th} = \frac{V_{oc}}{I_{sc}}$$

$$\text{KCL: } -2\angle 45 + \frac{V_1}{20} + \frac{V_1 - 10(\frac{V_1}{20})}{j10} = 0$$



$$\frac{V_1(1-j1)}{20} = 2\angle 45 \Rightarrow V_1 = \frac{40\angle 45}{\sqrt{1^2+1^2} \angle \tan^{-1} \frac{-1}{1}}$$

$$V_1 = \frac{40\angle 45}{\sqrt{2} \angle -45} = 20\sqrt{2} \angle 90$$

$$I_{sc} = \frac{10I_x}{10} = I_x = \frac{V_1}{20} = \frac{20\sqrt{2} \angle 90}{20} = \sqrt{2} \angle 90$$

$$Z_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{10\angle 45}{\sqrt{2} \angle 90} = 5\sqrt{2} \angle -45 = \boxed{5-j5 \Omega}$$

Thevenin equivalent circuit:

